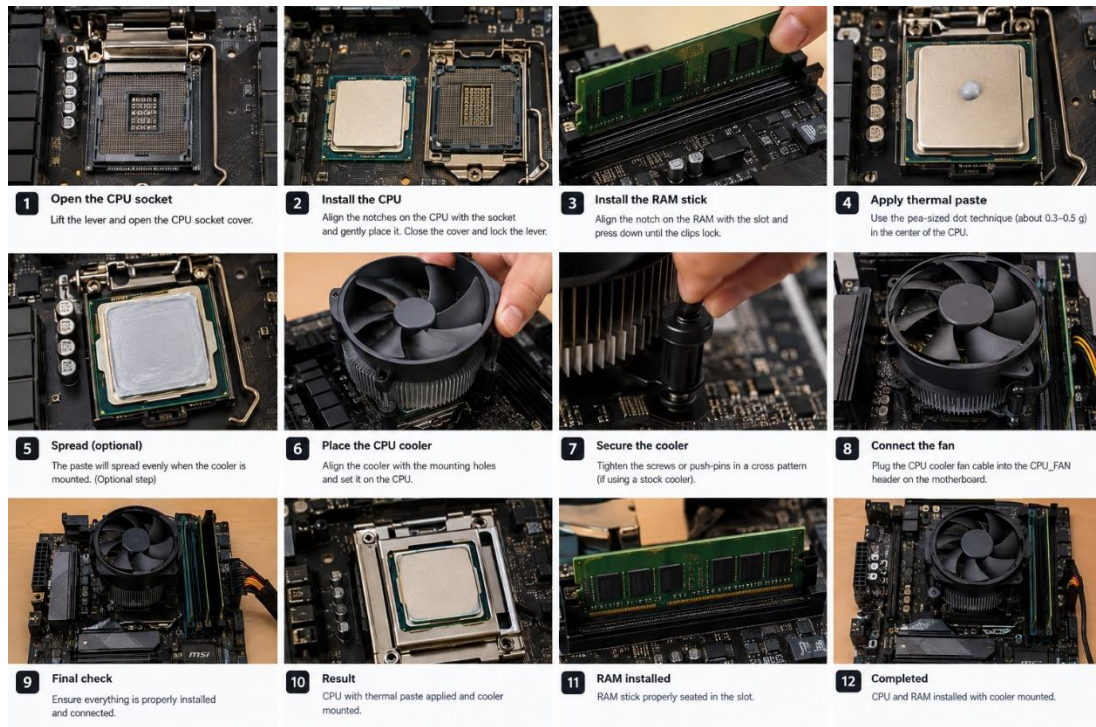


PC ASSEMBLY FROM COMPONENTS

1. Install a CPU and a RAM stick onto a motherboard, then apply thermal paste using the pea-sized dot technique before mounting the CPU cooler.

Hint: Take a clear photo of each step as you complete it for your documentation.

->



CPU and RAM Installation — Documentation

1. **Install the CPU:** Open the CPU socket, align the CPU's notches/marker with the socket, gently place it in position, and lock the socket.
2. **Install the RAM:** Align the notch on the RAM stick with the motherboard slot and press firmly until the retention clips lock.
3. **Apply thermal paste:** Put **one pea-sized dot (about 0.3–0.5 g)** of thermal paste in the center of the CPU. Do not spread it manually unless the cooler manufacturer's instructions require it.
4. **Mount the CPU cooler:** Position the cooler directly over the CPU and secure it using the supplied screws/push-pins, tightening them evenly in a cross pattern.
5. **Connect the cooler fan:** Plug the fan cable into the motherboard's **CPU_FAN** header.

6. **Final check:** Confirm that the CPU, RAM, cooler, and fan connection are properly seated and secured.

Photo documentation: The generated image above shows the requested installation steps, including the CPU, RAM, pea-sized thermal-paste application, and cooler mounting.

2. Mount the motherboard inside a PC case and connect both the 24-pin ATX and 8-pin CPU power connectors from the power supply, making sure each connector clicks into place.

-> **Motherboard Mounting and Power Connection**

1. **Prepare the case:** Switch off and unplug the power supply. Place the case on a stable, non-conductive surface.
2. **Install the I/O shield:** If the motherboard has a separate I/O shield, fit it into the rear opening of the case.
3. **Install the standoffs:** Make sure the motherboard standoffs are positioned to match the motherboard's mounting holes. **Do not mount the motherboard directly against the case.**
4. **Place the motherboard:** Carefully lower the motherboard into the case, aligning the rear I/O ports with the opening and the mounting holes with the standoffs.
5. **Secure the motherboard:** Use the appropriate **Phillips-head motherboard screws** to secure it to the standoffs. Tighten them firmly but don't overtighten.
6. **Connect the 24-pin ATX power:** Locate the large **24-pin ATX connector** from the PSU and plug it into the motherboard's 24-pin socket. Press until the **retaining clip clicks** into place.
7. **Connect the 8-pin CPU power:** Locate the **8-pin EPS/CPU power connector** (often 4+4) and connect it to the motherboard's CPU power socket near the processor. Press until its **retaining clip clicks** into place.
8. **Final check:** Gently verify that both connectors are fully seated and that their clips are locked.

Documentation

Component	Connector	Location	Securing mechanism
Motherboard	—	Case standoffs	Phillips-head screws
Main power	24-pin ATX	Motherboard edge	Retaining clip/click
CPU power	8-pin EPS/CPU	Near CPU socket	Retaining clip/click



1 Prepare the case

- Switch off and unplug the power supply.
- Place the case on a stable, non-conductive surface.



2 Install the I/O shield

- Fit the I/O shield into the rear opening of the case (if your motherboard has a separate shield).



3 Install the standoffs

- Make sure the standoffs are positioned to match the motherboard's mounting holes. Do not mount the motherboard directly against the case.



4 Place the motherboard

- Carefully lower the motherboard, aligning the rear I/O ports with the opening and the mounting holes with the standoffs.



5 Secure the motherboard

- Use Phillips-head screws to secure the motherboard to the standoffs. Tighten firmly but don't overtighten.



6 Connect the 24-pin ATX power

- Locate the large 24-pin ATX connector from the PSU and plug it into the motherboard. Press until the retaining clip clicks into place.



7 Connect the 8-pin CPU power

- Locate the 8-pin EPS/CPU power connector (often 4+4) and plug it into the socket near the CPU. Press until the retaining clip clicks into place.



8 Final check

- Verify that both the 24-pin ATX and 8-pin CPU connectors are fully seated and their clips are locked.



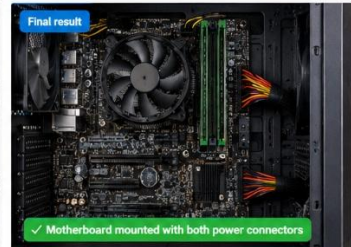
24-pin ATX connected

✓ Clicked into place



8-pin CPU power connected

✓ Clicked into place



Final result

✓ Motherboard mounted with both power connectors

3. Install a storage drive (SSD or HDD) and a GPU (graphics card) into the appropriate slots, then connect a power cables to each device.

->

1. Install the SSD

1. Install the case



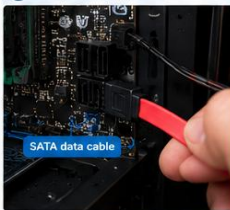
Attach the SSD to the 2.5" drive bracket using the provided screws.

2. Mount the SSD in the case



Slide the SSD with bracket into the 2.5" drive bay and secure it with screws (or tool-less clips, depending on the case).

3. Connect the SATA data cable



Plug one end of the SATA data cable into the SSD and the other end into an available SATA port on the motherboard.

4. Connect the SATA power cable



Connect a SATA power cable from the power supply to the SSD.

5. SSD installed and connected



The SSD is now **securely** mounted and both data and power cables are connected.

2. Install the GPU

1. Remove the slot cover



Remove the required PCIe slot cover from the case (usually 2 slots for a GPU).

2. Insert the GPU



Align the GPU with the PCIe x16 slot and gently press it in until it is fully seated.

3. Secure the GPU



Use the case screws to secure the GPU's rear bracket to the case.

4. Connect the PCIe power cable



Plug the PCIe power cable (6-pin or 8-pin) from the power supply into the GPU.

5. GPU installed and connected

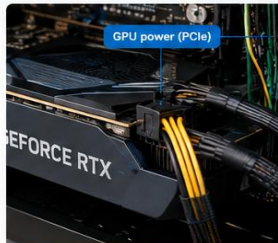


The GPU is now **securely** mounted and both data (PCIe slot) and power cables are connected.

3. Final look – All cables connected



SSD connected (SATA data + power)



GPU power connected (6/8-pin)



24-pin ATX and 8-pin CPU power connected



Final result – Everything installed and connected

4. Organize and secure all cables inside the case using zip ties or Velcro straps to ensure good airflow and a tidy appearance.

Hint: Take a photo showing your final cable management from both the front and back of

the case .



1 Identify and organize cables
Route all cables to their respective areas (right side of case, behind motherboard tray, and drive bays).



2 Use Velcro straps / zip ties
Bundle cables together using Velcro straps or zip ties for a clean and secure fit.



3 Route through cable channels
Use the case's cable management holes and channels to keep cables out of the airflow path.



4 Secure drive cables
Keep SATA data and power cables tidy and fastened near the drive bays using zip ties or Velcro.



5 Final check
Make sure all cables are properly secured, not loose, and do not block fans or airflow.



Cables used and secured

- 24-pin ATX power (motherboard)
- 8-pin CPU power
- PCIe power (GPU)
- SATA data (SSD/HDD)
- SATA power (SSD/HDD)
- Front panel / other cables

Benefits of proper cable management

- Improves airflow and cooling
- Keeps the build clean and organized
- Makes future upgrades easier
- Reduces cable damage and wear

5. Perform a final verification: check for any loose screws, confirm all power/data connections are firmly attached, and document your completed build with a series of labeled photos showing each major component installed.

->

✓

Final Verification Checklist

✓

No loose screws (case, motherboard, GPU, etc.)

✓

All power connections are firmly attached

✓

All data connections are secure

✓

Cables are neatly organized

✓

Fans are spinning freely

✓

System powers on and boots correctly

✓

Build Status: **COMPLETE**

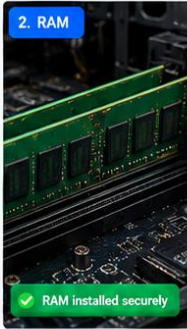
All components installed and verified!

1. CPU & Cooler



✓ CPU installed with cooler

2. RAM



✓ RAM installed securely

3. GPU



✓ Graphics card installed & powered

4. Storage (SSD)



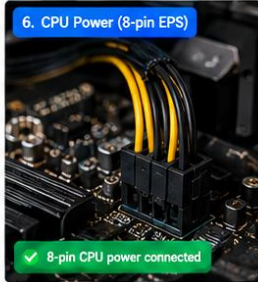
✓ SSD installed & connected

5. Motherboard Power (24-pin ATX)



✓ 24-pin ATX connected

6. CPU Power (8-pin EPS)



✓ 8-pin CPU power connected

7. SATA Power & Data (SSD)



✓ SATA power & data connected

8. PCIe Power (GPU)



✓ PCIe power connected

9. Case Fans



✓ Fans connected & working

10. Inside View (Side Panel Removed)



✓ All components installed and connected

11. Back View (Cable Management)



✓ Cables organized with Velcro straps

12. Final Build (Front View)



✓ Build completed successfully

Key Components Summary

 **CPU**
AMD Ryzen (or similar)

 **CPU Cooler**
Air Cooler (e.g., AMD stock / aftermarket)

 **Motherboard**
ATX / Micro-ATX

 **RAM**
DDR4 / DDR5 (e.g., 16GB)

 **GPU**
NVIDIA GeForce RTX (or similar)

 **Storage**
SSD (e.g., Samsung 1TB)

 **Power Supply**
550W (or as required)

 **Case**
Mid Tower (e.g., Cooler Master)

✓

PC Build Complete!
Ready for use.